MOHAMED NADIM TOUIL

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PROFILE

Robotics and AI software engineer specialized in autonomous systems, with hands-on experience in ROS 2, SLAM, LiDAR perception, and reinforcement learning for mobile robots. Strong background in embedded systems and sensor fusion, with international research experience in applied robotics.

EDUCATION

National Engineering Degree in Advanced Technologies 2021 – 2024
National School of Advanced Sciences and Technologies at Borj Cedria (ENSTAB), University of Carthage, Tunisia
Specialization: Embedded systems, electronics, control, and robotics

Preparatory Institute for Engineering Studies (IPEIM), University of Monastir, Tunisia 2019 – 2021
Intensive program in mathematics, physics, engineering sciences, and computer science
Prepared for the national competitive exam for engineering schools

SKILLS

Programming: Python, C, C++
Embedded/IoT: ESP32, STM32, Arduino, Raspberry Pi, Jetson, MQTT, LoRa
Tools: Linux, Git, Docker, ROS 2, Gazebo, SLAM, Mujoco
Protocols: UART, I2C, SPI

EXPERIENCE

Robotics & AI Software Engineer, EagleProjects Tunisia - Tunis, Tunisia May 2025 – Present
- Implemented SLAM and navigation pipelines in ROS 2 for Go2 quadruped robots
- Developed autonomous exploration and geofencing algorithms for Go2
- Developed and trained reinforcement learning controllers for locomotion motion control for G1 humanoid robots

ROS and Robotics Mentor (Part-time), Graine d'Entrepreneur - Tunis, Tunisia February – June 2025
Mentored interns on ROS 1/ROS 2, sensor integration, and autonomous system development.

Engineering Intern, VGR Laboratory, York University - Toronto, Canada May - August 2024
- Developed an autonomous surface vessel using ROS 2, integrating LiDAR and camera
- Implemented ArUco-based visual docking and trajectory planning for precise autonomous docking

Automation Intern, Aroteq - Ben Arrous, Tunisia June- August 2023
- Designed and programmed a 3-axis automated machine with Siemens S7-1200 PLC and SCADA interface.

End-of-Year Project, CMR Group - Ben Arrous, Tunisia January - May 2023
- Built an RFID-based system for cable harness tracking and real-time monitoring.

End-of-Year Project, LARINA Laboratory - Ben Arrous, Tunisia January - May 2022
- Designed an intelligent hydroponic system using Raspberry Pi, sensors, and automated nutrient control.

PROJECTS

Building Competitive Autonomous Robots 2022 – Present
Designed and programmed autonomous robots using PID control and sensor fusion.

Autonomous In-Office Delivery Robot 2024
Developed a small four-wheeled robot using ROS 2 for autonomous indoor delivery; implemented LiDAR-based obstacle avoidance, Nav2 path planning, and SLAM (gmapping) for safe navigation between waypoints.

IoT Health Monitoring System 2024
Built an ESP32 prototype with DHT22 and heart sensor, transmitting data via MQTT for anomaly detection and visualization.

Low-Power Edge IoT Pipeline 2023
Designed an IoT pipeline from ESP32 sensor nodes to a Raspberry Pi gateway and InfluxDB dashboard for scalable monitoring.

Eco-Energy Forest Fire Detection System 2023
Developed a wind-powered IoT-based fire detection system with GSM/email alerting.

Study of an Automatic Speed Management System 2022
Achieved 3rd place in a national hackathon; built an IoT + GPS based system for vehicle speed control and smart mobility.

LANGUAGES

English (Fluent) · French (Fluent) · Arabic (Native)

EXTRA-CURRICULAR ACTIVITIES

Mentoring, Hackathons, Student Robotics Club